

SA3050A – Business Intelligence & Visualization

Climate & Environmental Intelligence Dashboard

Advanced Power BI Solution for Global Environmental Monitoring

Group

7

The Business Problem

RISKS & VULNERABILITIES

Global Atmospheric Threat

Rapid climate shifts and escalating urban air pollution pose severe risks to:



Public Health: Rising respiratory diseases from airborne fine particulates.



Urban Infrastructure: Weather extremes pressuring city operations.



Sustainability: Long-term degradation of regional ecosystems.

INFORMATION GAP

Lack of Centralized BI

Environmental Protection Agencies, Health Departments, and Urban Policy Makers lack real-time integrated analytics, leading to:

- ✗ Inability to detect real-time pollution hotspots.
- ✗ Delayed monitoring of hazardous PM2.5 & PM10 levels.
- ✗ Reactive rather than proactive environmental policy.

Dataset Overview & Statistics

154,556

TOTAL OBSERVATIONS

41

DATA COLUMNS

2

RELATIONAL TABLES

Daily

UPDATE FREQUENCY

Source: Kaggle – Global Weather Repository

Daily meteorological and atmospheric records collected across world capital cities starting August 29, 2023.

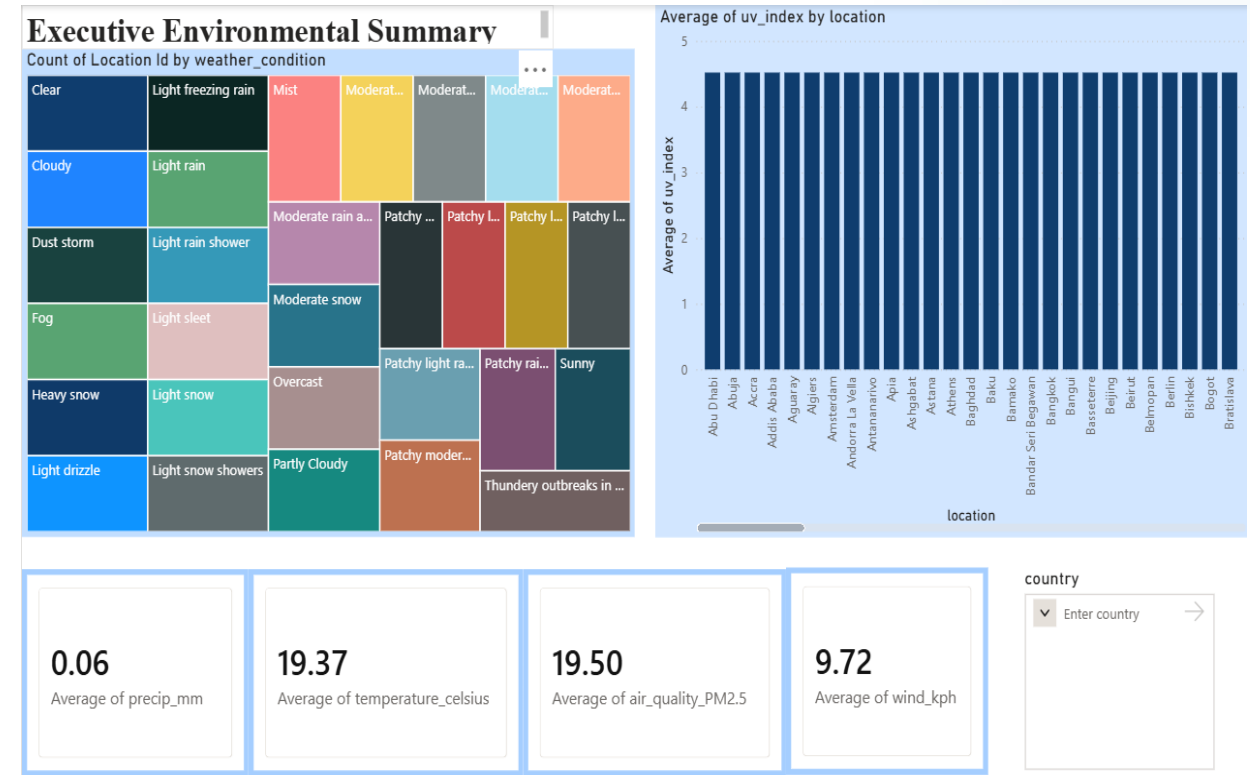
Weather: Temp, Rainfall, Humidity, Wind, Pressure, Visibility

Air Quality: PM2.5, PM10, Carbon Monoxide, UV Index

Scope: Global Capital Cities Climate Trend Analysis

🎯 Project Key Objectives

- 🖥️ **Real-Time Monitoring:** Track atmospheric and weather changes dynamically across global capital cities.
- 📈 **Air Quality Analytics:** Evaluate PM2.5, PM10, and Carbon Monoxide pollution profiles.
- 🌐 **Geographical Comparison:** Benchmark environmental indicators between countries and cities.
- 🔔 **Risk Detection:** Automatically pinpoint extreme weather events and critical health hazard hotspots.
- ⚖️ **Evidence-Based Policy:** Empower decision-makers with actionable, data-backed insights.



? Core Business Questions



Pollution Hotspots

Which regions & capital cities record the highest pollution levels and critical PM2.5 concentrations?



Climate Patterns

How do temperature, rainfall, and humidity fluctuate globally over time across reporting periods?



Weather Correlations

Are there direct relationships between local meteorological factors and worsening air quality metrics?



Precipitation Distribution

Which geographic locations receive extreme rainfall vs prolonged dry spells?



Critical Alerts

Which urban locations require immediate targeted environmental intervention and public warnings?



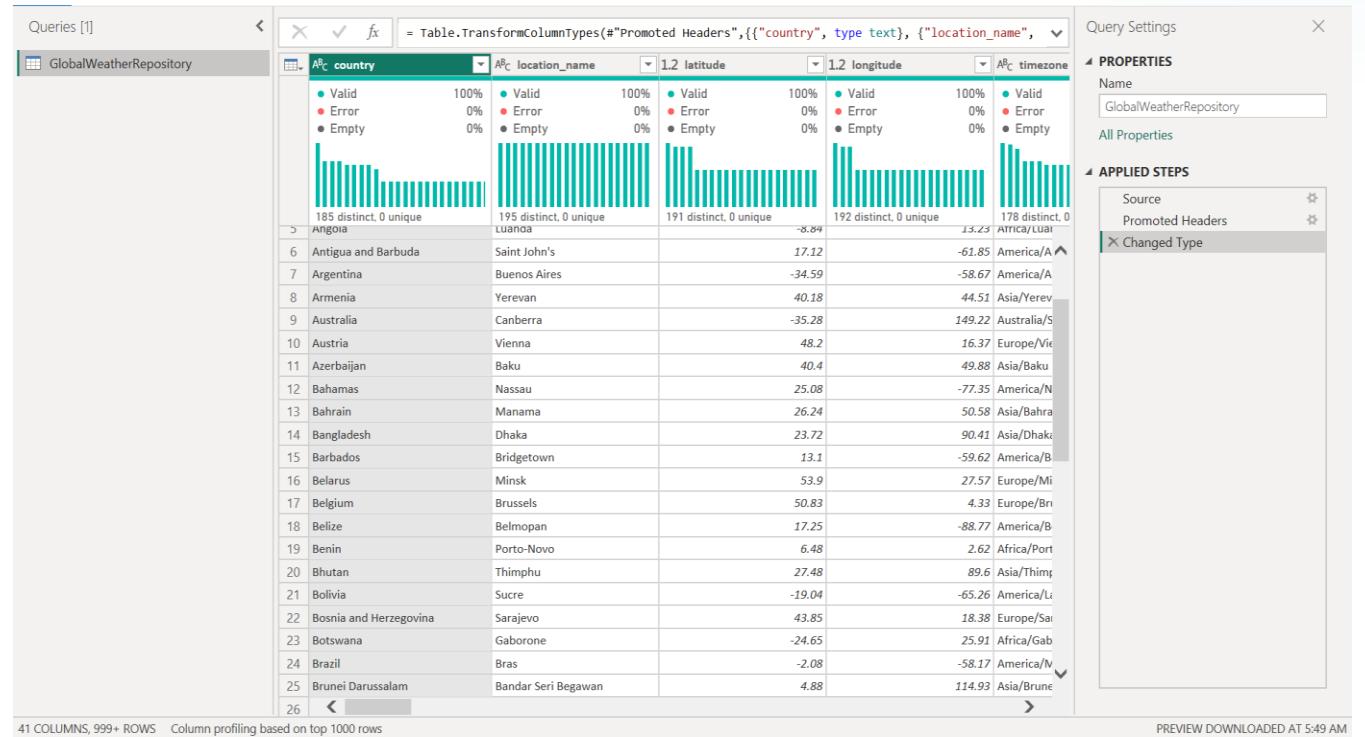
Policy Support

How can integrated BI metrics guide urban policy planning and emission control strategies?

Power Query Data Cleaning

ETL Transformation Workflow

- ✓ Standardized column names & set correct data types.
- ✓ Removed duplicate rows, blank entries & invalid values.
- ✓ Cleaned text fields & merged geographic attributes.
- ✓ Handled missing metrics using contextual imputations.
- ✓ Built custom Date Table & extracted Year/Month/Date.
- ✓ Created dedicated Regional Summary Table for aggregations.



Star Schema Data Model

Fact Table

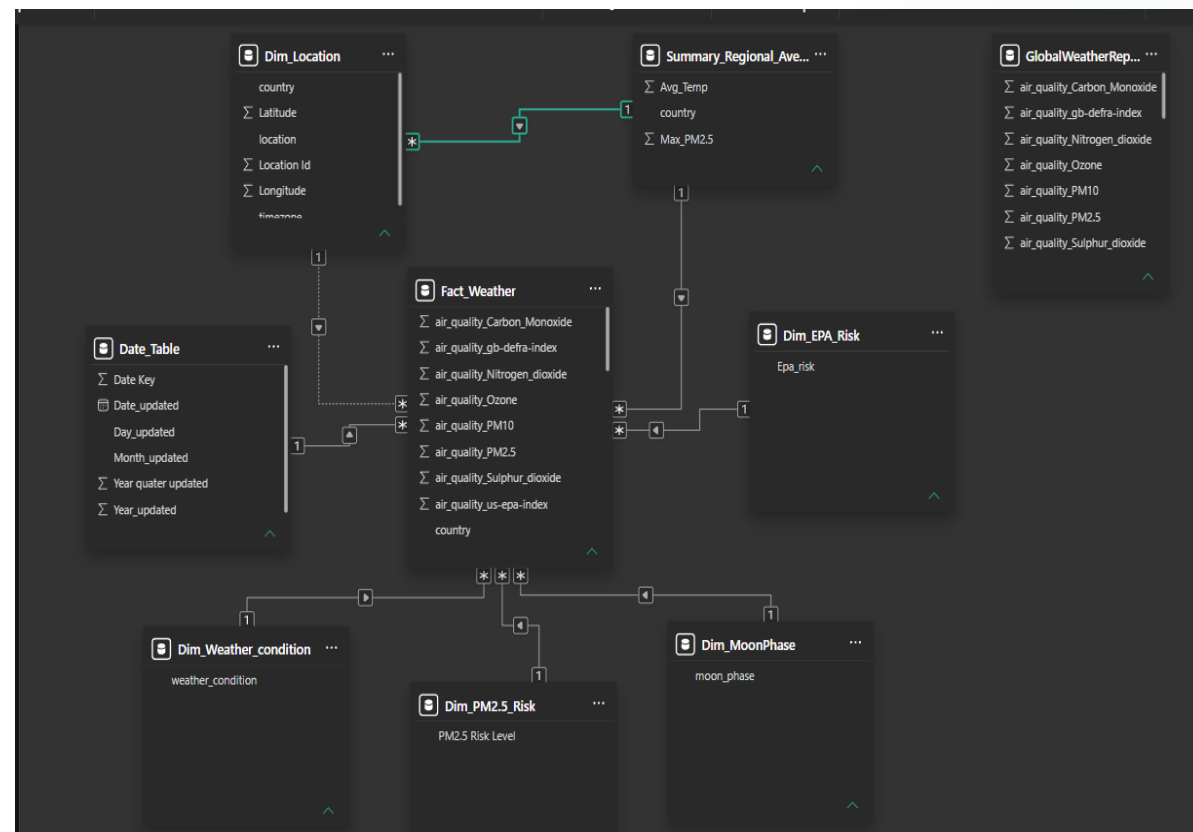
Weather Observations: Central facts containing daily weather metrics, particulate measurements, and atmospheric indicators.

Dimension Tables

Date Dim | Country Dim | Location Dim | Weather Dim | PM2.5 Risk Dim | Moon Phase Dim | Regional Summary

Model Benefits:

High query speed, time intelligence calculations, reduced redundancy & seamless UI slicing.



Key DAX Measures Created



KPI Measures

Avg Temperature
Avg Humidity
Avg UV Index
Total Rainfall



Ranking Measures

Dynamic location rankings for:

- Temperature Rank
- Rainfall Rank
- PM2.5 Severity Rank



Ratio & Status

- PM2.5 Ratio
- Visibility Ratio
- Air Quality Status
- Heat Alert Indicator



Dynamic Titles

Slicer-driven dynamic metrics:

- Selected Weather
- Dynamic Dashboard Header
- Contextual Tooltips

Dashboard Report Structure

PAGE 1

Executive View

High-level KPIs summarizing global temperature, humidity, total rainfall, UV index, air quality status, and extreme heat alert indicators.

PAGE 2

Air Quality

Deep dive into hazardous particulate matter: PM2.5, PM10, Carbon Monoxide concentrations, regional rankings & risk categories.

PAGE 3

Climate Trends

Time-series intelligence illustrating multi-year temperature evolution, rainfall patterns, barometric pressure & visibility metrics.

PAGE 4

Regional Analysis

Interactive geographic matrix facilitating direct country-to-country and city-to-city comparisons across environmental indicators.

💡 Major Findings & Insights

19.50

GLOBAL AVG PM2.5 (MODERATE RISK)

20.21°C

GLOBAL AVERAGE TEMPERATURE

- ⚠️ **High Carbon Monoxide Hotspots:** Consistently elevated levels in cities including *Abu Dhabi, Abuja, Accra, Addis Ababa, and Aguaray*.
- 📈 **Temperature Stability:** Global temperatures remained relatively stable between 2024 and 2026.
- 🌙 **Moon Phase Distribution:** Waxing Gibbous accounts for ~23% of recorded observations.
- 🔄 **Pressure vs Emissions:** Atmospheric pressure remained stable, proving that local air quality changes are driven by pollution emissions rather than pressure shifts.

Strategic Recommendations



1. Emissions Policy

Tighten vehicle emission standards and enact strict industrial release limits in urban centers.



2. Clean Transit

Expand green electric public transit networks to diminish carbon monoxide reliance.



3. Expand Stations

Increase air quality monitoring stations across densely populated capital cities.

4. Public Awareness Campaigns

Issue timely health warnings and public advisory alerts during poor air quality spells.

5. Long-term Climate Tracking

Sustain daily monitoring of pressure and temperature metrics to detect long-term climate shifts.



Conclusion & Project Impact

The **Climate & Environmental Intelligence Dashboard** successfully transforms millions of complex raw observations into an interactive, high-impact Business Intelligence platform. It empowers stakeholders to detect pollution hotspots, benchmark climate trends, and drive evidence-based environmental policy worldwide.